

Fahimeh Baftizadeh

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Summary — Computational Scientist with 10+ years of experience applying large-scale modeling and machine learning to complex biological systems. During my PhD at SISSA, I developed molecular simulations of peptide aggregation relevant to neurodegenerative diseases, followed by postdoctoral research at MIT where I collaborated with Novartis on computational modeling of drug crystallization. At the Allen Institute for Brain Science, I modeled the effects of extracellular electric fields on neuronal activity and later transitioned to a Machine Learning Scientist role, where I developed ML models to analyze large-scale genomics, electrophysiology, and morphology datasets.

I am expert in Python, PyTorch, and GPU/CPU cluster computing (thousands of simulations executed). I have independently built custom AI agents (LangChain, Cohere) and implemented deep learning architectures including GPT-style models. I enjoy working at the intersection of science, engineering, and customer engagement, translating complex technical problems into scalable AI solutions that deliver real-world value.

Experience

Career Break — Global Travel & Independent ML/AI Projects & Volunteer Work

2024 – Present

Planned career break to pursue long-term travel

- Traveled for 13 months, visiting 25 countries across 5 continents.
- Built deep learning projects including a Variational Autoencoder (VAE) on 500K+ recipes for similarity recommendations ([See here](#)), CNNs, RNNs ([See here](#)), and a small GPT-style model ([See here](#)) to gain hands-on experience with modern architectures .
- Volunteered at Golestan Foundation to improve operational workflows: Identified repetitive CSV consolidation bottleneck (2-3 hours/cycle) and built an AI agent to automate data cleaning, handle natural-language instructions, and generate import-ready datasets—reducing processing time to minutes. ([See here](#)).
- Diagnosed inefficient task management processes and developed a lightweight tool to assign tasks, track progress, and automate reminders—streamlining team workflow and accountability. ([See here](#)).

Allen Institute for Brain Science

2017 – 2024

Machine Learning Scientist II

- Developed a coupled-autoencoder machine-learning algorithm to integrate and analyze multimodal single-cell data, enabling prediction of missing measurements and improving clustering of cellular populations. Contributed to an institute-wide initiative on multimodal data analysis ([See here](#)).
- Designed a KL-divergence-based metric to quantify data quality across heterogeneous single-cell platforms (Patch-seq, FACS), standardizing cross-experiment measurement consistency.

Computational Scientist

- Modeled extracellular electric field effects on cortical neurons, revealing cell-class-dependent entrainment patterns that enable selective neuromodulation design.
- Collaborated with engineers to extend an internal Python analysis package, adding new features to support evolving single-cell research needs and improve usability across teams.

Massachusetts Institute of Technology

2012 – 2016

Postdoctoral Researcher

- Developed computational method in collaboration with Novartis to predict optimal polymer surface nano-patterns for directed drug crystallization through MD simulations.
- Executed over 500,000 high-resolution simulations (30,000 particles each) on HPC clusters, generating terabytes of data for downstream analysis.

- Simulated formation of ordered amyloid-like structures from disordered peptide aggregates using bias-exchange metadynamics and MD simulations, revealing a complex nucleation pathway.
- Processed and analyzed long time-series simulation data using advanced statistical methods, including clustering and Weighted Histogram Analysis Method (WHAM).

Education

Massachusetts Institute of Technology

Nov 2012 – Aug 2016

Postdoctoral Researcher

SISSA, the International School for Advanced Studies, Trieste, Italy

Oct 2008 – Oct 2012

Ph.D., Physics and Chemistry of Biological Systems

Ferdowsi University of Mashhad, Iran

Sep 2002 – Sep 2007

Master's and Bachelor's Degrees, Physics

Skills

Machine Learning & AI

- Python, PyTorch, Scikit-learn, NumPy, Pandas

Scientific Computing & Simulation

- Molecular Dynamics: GROMACS, VMD
- Computational Neuroscience: NEURON, BioNet

High-Performance Computing

- CPU/GPU clusters, distributed job management, Linux/bash scripting

Selected Publications

20+ peer-reviewed publications in high-profile journals, including Nature, Cell, and Science. ([See here](#))

1. Y. Marghi, R. Gala, **F. Baftizadeh**, U. Sümbül, “Joint inference of discrete cell types and continuous type-specific variability in single-cell datasets with MMIDAS”, *Nature Computational Science*, 706-722
2. S. Y. Lee, K. Kozalakis, **F. Baftizadeh**, et. al., “Cell-class-specific electric field entrainment of neural activity”, *Neuron*, P2614-2630.E5
3. R. Gala, A. Budzillo, **F. Baftizadeh**, et al., “Consistent cross-modal identification of cortical neurons with coupled autoencoders”, *Nature Computational Science*, 120-127
4. Z. Yao, C. Velthoven, T. Nguyen, J. Goldy, A. Cortes, **F. Baftizadeh**, et. al., “A taxonomy of transcriptomic cell types across the isocortex and hippocampal formation”, *Cell*, P3222-3241.E26
5. N. Gouwens, S. Sorensen¹, **F. Baftizadeh**, et al., “Integrated Morphoelectric and Transcriptomic Classification of Cortical GABAergic Cells”, *Cell*, P935-953.E19
6. **F. Baftizadeh**, X. Biarnex, F. Pietrucci, F. Affinito, A. Laio, “Multidimensional View of Amyloid Fibril Nucleation in Atomistic Detail”, *Journal of American Chemical Society*, 3613-3614
7. **F. Baftizadeh**, P. Cossio, F. Pietrucci, A. Laio, “Protein Folding and Ligand-Enzyme Binding from Bias-Exchange Metadynamics Simulations”, *Current Physical Chemistry*, 79-91
8. **F. Baftizadeh**, F. Pietrucci, X. Biarnes, A. Laio, “Nucleation Process of a Fibril Precursor in the C-Terminal Segment of Amyloid-beta”, *Physical Review Letter*, 168103
9. D. Garanata, **F. Baftizadeh**, et. al., “The inverted free energy landscape of an intrinsically disordered peptide by simulations and experiments”, *Scientific Reports*, 15449